

Technology Stack

Date	08 October 2026
Team ID	UNM1611PUNITHA
Project Name	AI StudyBuddy API
Maximum Marks	

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Frontend / Client-Side	HTML, CSS, JavaScript	Used to build the user interface for uploading study materials, viewing summaries, flashcards, quizzes, and study plans. These technologies provide a simple and responsive client-side interface.
1			Priority: High

2	Backend / Server-Side	Node.js + Express.js	Used to build the RESTful API and handle authentication, study-material management, AI requests, and communication between the frontend, database, and Gemini AI. Priority: High
3	Database / Data Storage	MongoDB	Used to store user accounts, authentication-related data, uploaded study-material information, summaries, flashcards, quizzes, and study plans. MongoDB is suitable for flexible document-based data. Priority: High
4	Cloud / Hosting / Deployment	Render	Used as a simple cloud hosting option for deploying the Node.js and Express.js backend API. It provides an easy deployment workflow suitable for a student project prototype. Priority: Medium
5	Version Control & CI/CD	Git + GitHub	Git is used for version control and tracking project changes. GitHub is used to store the source code, collaborate with team members, and maintain the project repository. Priority: High

	Collaborative repository management and project history.		
6	Third-Party APIs / Other Tools	Google Gemini API + Postman + MongoDB Compass	Google Gemini API enables AI-powered generation of summaries, flashcards, quizzes, and personalized study plans. Postman is used to test and validate REST API endpoints. MongoDB Compass is used to view and manage MongoDB data during development. Priority: High